Metals Market Index MMI

MMi Dashboard

Iron Ore Price Indices

Iron Ore Port Stock (FOT Qingdao)
IOPI62 62% Fe Fines RMB/t

858
-34 -3.81%
Apr 21st, 2023

Iron Ore Port Stock (FOT Qingdao)
IOPI65 65% Fe Fines RMB/t

970
-34.00 -3.39%
Apr 21st, 2023

Iron Port Stock (FOT Qingdao)
IOPI58 58% Fe Fines RMB/t

794
1 0.13%
Apr 21st, 2023

Iron Ore Seaborne (CFR Qingdao)
IOSI62 62% Fe Fines USD/dmt

145.75
0.00 0.00%
Apr 21st, 2023

Iron Ore Seaborne (CFR Qingdao)
IOSI65 65% Fe Fines USD/dmt

149.10
0.00 0.00%
Apr 21st, 2023

Iron Ore Port Stock (FOT Qingdao)
IOPLI 62.5% Fe Lump RMB/t

975
-35 -3.47%
Week Ending Apr 21st, 2023

Exchange Traded Contracts

DCE Iron Ore 62% Fines
I2309 (Jan) RMB/t (3pm close)

730.50
-31.00 -4.07%
Apr 21st, 2023 (3pm close)

SGX Iron Ore (CFR Qingdao) 62% Fe Fines
May 23 USD/dmt

108.00
-7.45 -6.45%
Apr 21st, 2023 (5.30 pm Print)

SHFE Rebar
RB2310 (Jan) RMB/t

3810
-104 -2.66%
Apr 21st, 2023 (3pm close)

Freight Rates

C3, Tubarao - Qingdao USD/t

21.39
0.21 0.97%
Apr 20th, 2023

C5, W. Australia - Qingdao USD/t

8.07
0.36 4.67%
Apr 20th, 2023

Steel Price

Steel Rebar (China Domestic) RMB/t

3970
-90 -2.22%
Week Ending Apr 21st, 2023

Inventory Levels

Iron Ore Inventory at Chinese Ports (35)
million tonnes

126.95
1.18 0.94%
Week Ending Apr 21st, 2023

Steel Inventory in China
million tonnes

16.39
-0.47 -2.80%
Week Ending Apr 21st, 2023

Steel Price

Steel HRC (China Domestic) RMB/t

4140
-60 -1.43%
Week Ending Apr 21st, 2023

IRON ORE PORT STOCK INDEX (IOPI)

Apr 21st, 2023		FOT Qingdao (inc. 13% VAT), RMB/wet tonne							CFR Qingdao Equivalent (exc. 13% VAT), USD/dry tonne [1]						
Index	Fe Content	Price	Change	Change %	MTD	YTD	Low [2]	High [2]	Price	Change	Change %	MTD	YTD	Low [2]	High [2]
IOPI62	62% Fe Fines	858	-34	-3.8%	869	880	858	892	115.73	-4.74	-3.9%	120.06	120.58	117.93	122.93
IOPI58	58% Fe Fines	794	1	0.1%	813	798	761	793	107.98	0.14	0.1%	113.29	110.17	105.29	110.04
IOPI65	65% Fe Fines	970	-34	-3.4%	981	992	970	1003	131.35	-4.74	-3.5%	136.05	136.43	133.84	138.73

IRON ORE SEABORNE INDEX (IOSI)

Apr 21st, 2023		CFR Qingdao, USD/dry tonne						
Index	Fe Content	Price	Change	Change %	MTD	YTD	Low [2]	High [2]
IOSI62	62% Fe Fines	145.75	0.00	0.00%	114.38	124.32	83.90	149.05
IOSI65	65% Fe Fines	149.10	0.00	0.00%	124.18	140.25	94.45	173.75

MARKET COMMENTARY

DCE iron ore futures declined by 4.82% today, the main contract closed at 730.5.The traders' willingness to ship is better.The steel mills are active to purchase.The overall trading sentiment of the market is better.PBF at Shandong port deal 847-856 yuan/mt,decrease 19-21 yuan/mt; PBF at Tangshan port deal 843-860 yuan/mt,decrease 27-29 yuan/mt. As of April 21st, SMM has tracked a total of 126.95 million tons of inventory in 35 ports, a cumulative of 1.17 million tons compared to last week. The daily average port clearance volume of imported ore in this period increased by 13000 tons to 3.1 million tons on a weekly basis. The daily average port clearance volume was once again supported by the high production of blast furnace hot metal from steel mills this week and the replenishment of warehouses before May Day. There will still be some restocking demand in the market next week, and it is expected that the decline in port clearance will be relatively slow. However, inventory at Port 35 may accumulate again next week. Considering the impact of the Federal Reserve's interest rate hike and demand data for building materials, there may be a slight rebound in mining prices next week, but the overall trend is still dominated by broad fluctuations.

IRON ORE PORT LUMP INDEX (IOPLI)

Week Ending Apr 21st, 2023		FOT Qingdao (inc. 13% VAT), RMB/wet tonne							CFR Qingdao Equivalent (exc. 13% VAT), USD/dry tonne [3]						
Index	Fe Content	Price	W-o-W	Change %	MTD	YTD	Low [2]	High [2]	Price	W-o-W	Change %	MTD	YTD	Low [2]	High [2]
IOPLI62	62.5% Fe Lump	975	-35	-3.5%	926	1031	770	1330	126.81	-10.45	-7.61%	119.59	137.82	95.00	183.00

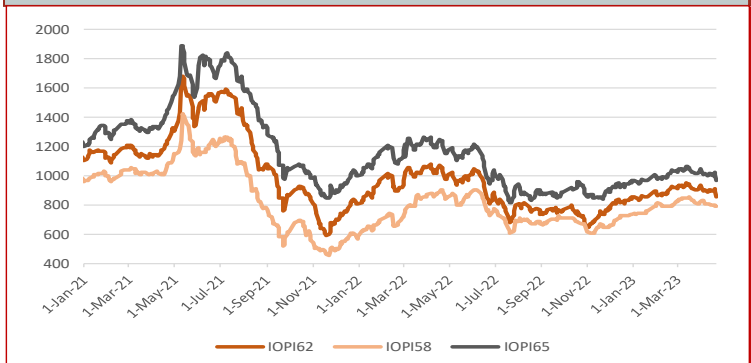
IRON ORE DOMESTIC CONCENTRATE SPOT PRICE ASSESSMENTS AND COMPOSITE INDEX

Week Ending Apr 21st, 2023				RMB/tonne (excluding tax) [3]				USD/tonne (excluding tax) [3]			
Province	Region	Product	Basis	This week	Change %	Low [2]	High [2]	This week	Change %	Low [2]	High [2]
Hebei	Hanxing	66% Fe Concentrate	Dry	1019	-0.6%	779	1645	148.13	-0.64%	110.31	255.69
Hebei	Qian'an	65% Fe Concentrate	Dry	1035	-2.8%	780	1630	150.45	-2.87%	110.51	251.57
Liaoning	Anshan	65% Fe Concentrate	Wet	795	-1.2%	620	1310	115.56	-1.30%	87.40	202.32
Shandong	Zibo	65% Fe Concentrate	Dry	1034	-0.7%	800	1752	150.31	-0.73%	117.19	272.32

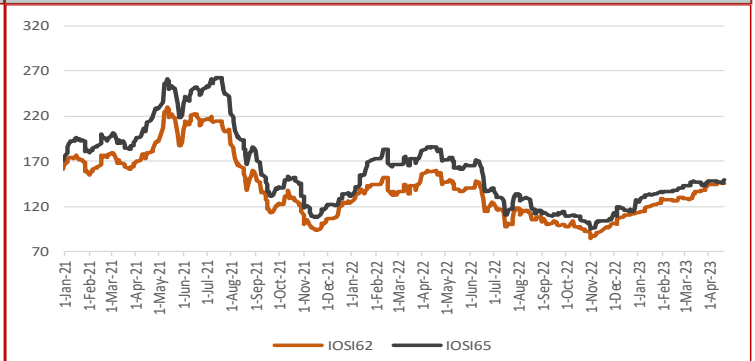
Week Ending Apr 21st, 2023	This week	Change %	Low [2]	High [2]
China Mines Concentrate Composite Index RMB/WT	927.13	-1.56%	706.36	1511.22

[1] Exchange rate applied: RMB/USD = 6.7522 [2] Last 12 months
[3] Weekly exchange rate applied: RMB/USD =6.87528

IRON ORE PORT INDEX, FOT QINGDAO (RMB/WT)



IRON ORE SEABORNE INDEX, CFR QINGDAO (USD/DMT)



IRON ORE PORT STOCK INDEX MONTHLY, QUARTERLY AND YEAR-TO-DATE AVERAGES

Apr 21st, 2023		FOT Qingdao (inc. 13% VAT), RMB/wet tonne							CFR Qingdao Equivalent (exc. 13% VAT), USD/dry tonne						
Index	Fe Content	December	January	February	March	MTD	QTD	YTD	December	January	February	March	MTD	QTD	YTD
IOPI62	62% Fe Fines	822	856	892	924	869	881	880	109.43	117.00	121.64	124.91	120.06	120.90	120.58
IOPI58	58% Fe Fines	711	751	799	832	813	792	798	95.17	103.34	109.79	113.41	113.29	109.35	110.17
IOPI65	65% Fe Fines	935	968	1003	1036	981	993	992	124.90	132.80	137.36	140.49	136.05	136.72	136.43

IRON ORE SEABORNE INDEX MONTHLY, QUARTERLY AND YEAR-TO-DATE AVERAGES

Apr 21st, 2023		CFR Qingdao, USD/dry tonne						
Index	Fe Content	December	January	February	March	MTD	QTD	YTD
IOSI62	62% Fe Fines	108.92	119.76	127.73	134.77	112.14	113.96	124.32
IOSI65	65% Fe Fines	116.83	132.39	138.06	145.14	122.86	124.22	140.25

FREIGHT RATES

Apr 20th, 2023		FREIGHT RATES - DRY BULK US$/wet tonne				
Route	Designation		Change	Change %	Low [2]	High [2]
W. Australia - Qingdao	C5	8.07	0.36	4.67%	3.57	16.77
Tubarao - Qingdao	C3	21.39	0.21	0.97%	6.70	36.40

IRON ORE PORT LUMP INDEX MONTHLY, QUARTERLY AND YEAR-TO-DATE AVERAGES

Week Ending Apr 21st, 2023		FOT Qingdao (inc. 13% VAT), RMB/wet tonne							CFR Qingdao Equivalent (exc. 13% VAT), USD/dry tonne [1]						
Index	Fe Content	December	January	February	March	MTD	QTD	YTD	December	January	February	March	MTD	QTD	YTD
IOPLI62	62.5% Fe Lump	922	964	988	1012	925	924	1032	117.06	126.47	129.95	131.15	119.39	119.69	137.99

IRON ORE INDEX PREMIUMS/DISCOUNTS

Apr 21st, 2023		PORT STOCK INDEX (RMB/WT)	
Index	Fe Content	Spread to IOPI62	% Spread to IOPI62
IOPI58	58% Fe Fines	-64	-7.46%
IOPI65	65% Fe Fines	112	13.05%

Apr 21st, 2023		SEABORNE INDEX (USD/DMT)	
Index	Fe Content	Spread to IOSI62	% Spread to IOSI62
IOSI65	65% Fe Fines	3.35	2.30%

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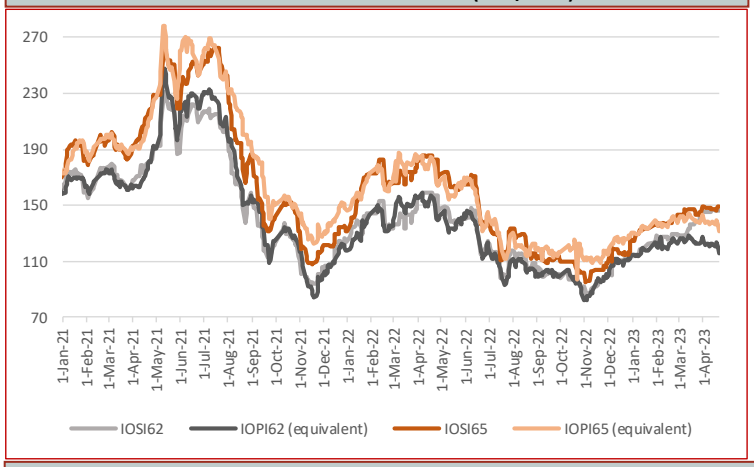
IRON ORE INDEX COMPARISONS (USD/DMT)



CHINA DOMESTIC COMPOSITE MINES INDEX (RMB/Wet Tonne, including VAT)

IRON ORE BRAND SPOT PRICE ASSESMENTS

Apr 21st, 2023	PORT STOCK INDEX (RMB/WT)		
	Price	Change	Diff to IOPI62
Roy Hill	801	-34	-57
SIMEC Fines	730	-35	-128
PB Fines	826	-34	-32
Newman Fines	827	-34	-31
MAC Fines	808	-34	-50
Jimblebar Blended Fines	723	-34	-135
Carajas Fines	1038	-35	180
Brazilian SSF	822	-34	-36
Brazilian Blend Fines	840	-33	-18
RTX Fines	741	-34	-117
West Pilbara Fines	769	-34	-89

Apr 21st, 2023	SEABORNE INDEX (USD/DMT)		
	Price	Change	Diff to IOSI62
Roy Hill	141.25	0.00	-4.50
SIMEC Fines	137.75	0.00	-8.00
PB Fines	142.00	0.00	-3.75
Newman Fines	144.90	0.00	-0.85
MAC Fines	142.00	0.00	-3.75
Jimblebar Blended Fines	134.40	0.00	-11.35
Carajas Fines	175.30	0.00	29.55
Brazilian SSF	149.50	0.00	3.75
Brazilian Blend Fines	151.15	0.00	5.40
RTX Fines	135.65	0.00	-10.10
West Pilbara Fines	140.00	0.00	-5.75

Apr 21st, 2023	PORT STOCK INDEX (RMB/WT)		
	Price	Change	Diff to IOPI58
SSF	794	1	0
FMG Blended Fines	804	1	10
Robe River	805	1	11
Western Fines	808	1	14
Atlas Fines	801	1	7
Yandi	787	1	-7

IRON ORE INDEX NORMALISATION DIFFERENTIALS

Port Stock Index Product Differentials (RMB/wet tonne)			
	Applicable range	Value	Change
1% Fe	High Grade Fe 60 - 63%	5.00	0.00
	High Grade Fe 63 - 64%	5.00	0.00
	High Grade Fe 64 - 65%	5.00	0.00
	High Grade Fe 65 - 65.5%	5.00	0.00
	Low Grade Fe	20.00	1.00
1% Alumina	High Fe Grade Al <2.25%	41.00	0.00
	High Fe Grade Al 2.25-4%	5.00	0.00
	Low Fe Grade Al <2.25%	82.00	-8.00
	Low Fe Grade Al 2.25-4%	18.00	0.00
1% Silica	High Fe Grade Si <4%	13.00	0.00
	High Fe Grade Si 4-6.5%	13.00	-1.00
	Low Fe Grade	17.00	0.00
0.01% Phosphorus	High Fe Grade 0.09%<P<0.115%	9.00	0.00
	High Fe Grade 0.115%<P<0.15%	9.00	0.00
	Low Fe Grade 0.09<P<0.1%	5.00	0.00

Seaborne Index Product Differentials (USD/dry tonne)			
	Applicable range	Value	Change
1% Fe	High Grade Fe 60 - 63%	1.25	-0.75
	High Grade Fe 63 - 64%	2.75	0.75
	High Grade Fe 64 - 65%	2.75	0.75
	High Grade Fe 65 - 65.5%	2.75	0.75
1% Alumina	High Fe Grade Al <2.25%	0.25	0.00
	High Fe Grade Al 2.25-4%	0.25	0.00
1% Silica	High Fe Grade Si <4%	1.75	0.25
	High Fe Grade Si 4 - 6.5%	4.25	0.00
0.01% Phosphorus	High Fe Grade 0.09%<P<0.115%	0.75	0.25
	High Fe Grade 0.115%<P<0.15%	7.25	0.75

Port Stock Price Differentials to Qingdao Port for PB Fines (RMB/wet tonne)

Port	Value	Change	Port	Value	Change	Port	Value	Change	Port	Value	Change
Bayuquan	-100.00	0.00	Fangcheng	-25.00	0.00	Lanshan	0.00	0.00	Rizhao	0.00	0.00
Beilun	0.00	0.00	Jiangyin	-30.00	0.00	Lianyungang	0.00	0.00	Shekou	0.00	0.00
Caofeidian	-10.00	0.00	Jingtang	-10.00	0.00	Majishan	0.00	0.00	Taicang	-30.00	0.00
Dalian	0.00	0.00	Lanqiao	0.00	0.00	Qingdao	0.00	0.00	Tianjin	-15.00	0.00

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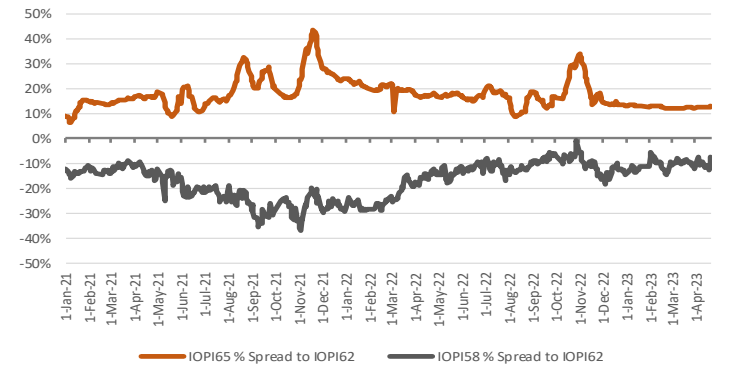
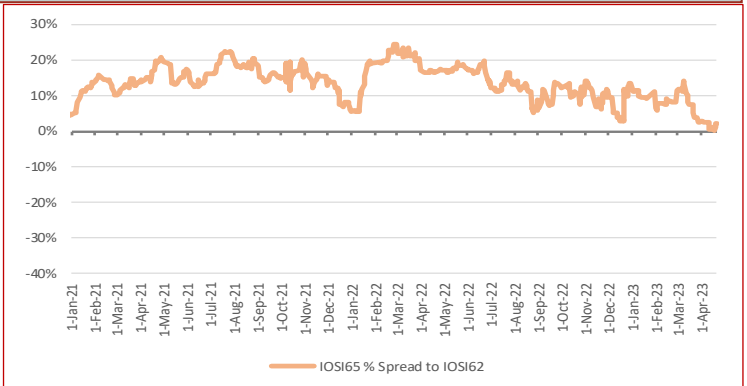
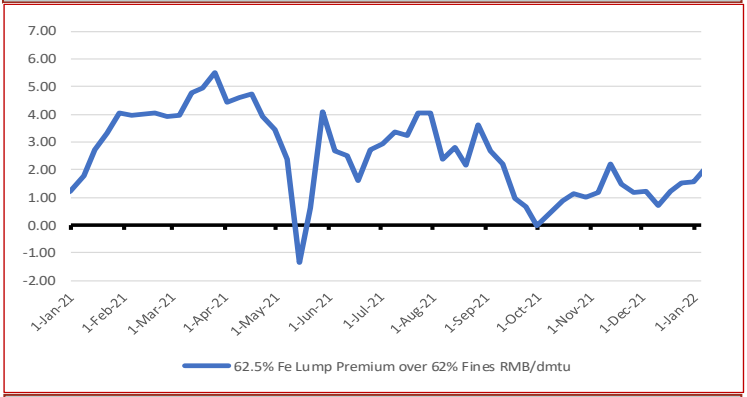
IRON ORE INDEX PREMIUMS/DISCOUNTS

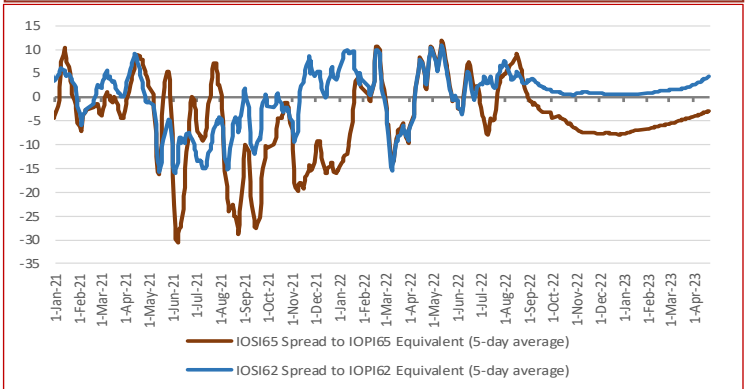




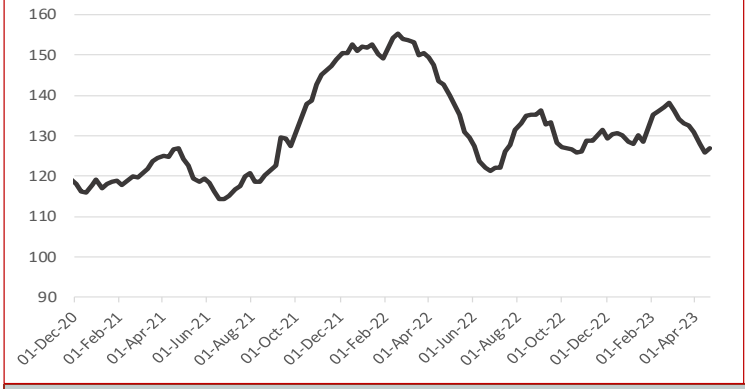
WEEKLY IRON ORE PORT STOCK LUMP PREMIUM (RMB/DMTU)



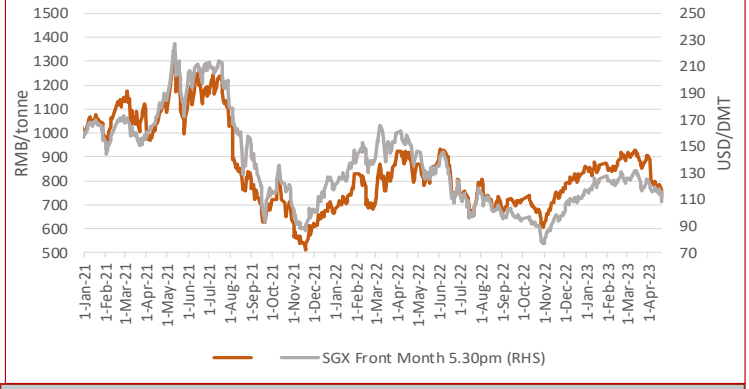
IRON ORE SEABORNE TO PORT STOCK PRICE SPREADS (USD/DMT)



TOTAL IRON ORE INVENTORIES AT CHINA PORTS (MILLION TONNES)



FUTURE TRADING—FRONT MONTH CLOSING PRICE



IRON ORE PORT INVENTORIES (MILLION TONNES)

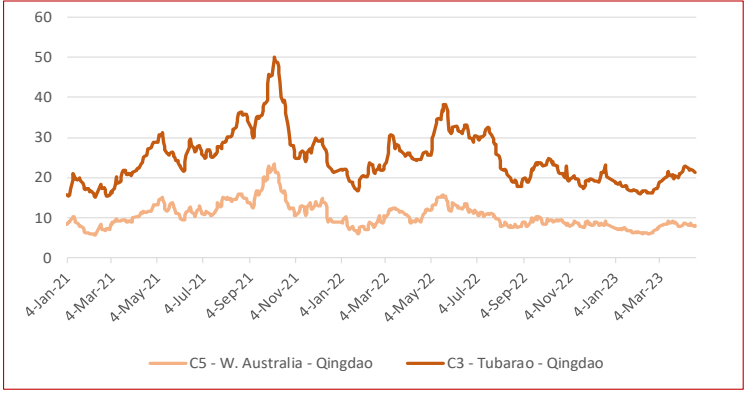
Week Ending Apr 21st, 2023

Province	This week	Change %	Low [2]	High [2]
Jingtang	12.23	-2.86%	9.84	19.20
Qingdao	25.10	2.24%	9.41	26.24
Caofeidian	12.98	4.59%	9.05	16.29
Tianjin	9.23	4.53%	7.14	12.97
Rizhao	16.88	-0.88%	9.44	19.26
Total (35 Ports)	126.95	0.94%	98.80	155.39

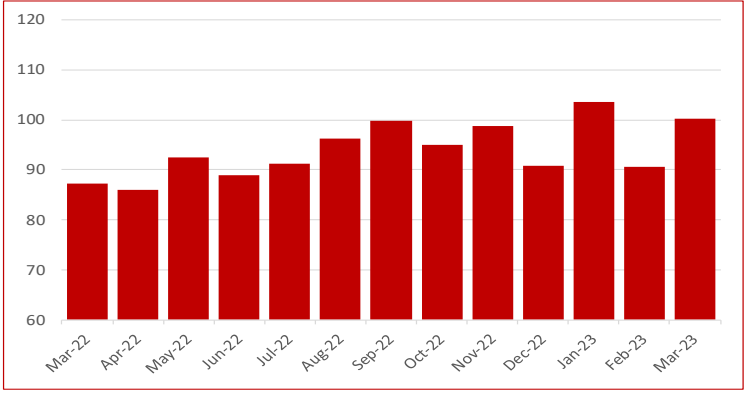
IRON ORE FUTURES CONTRACTS

	DCE (RMB/WMT)			SGX (USD/DMT)		
Closing Date	Apr 21st, 3pm close			Apr 21st, 5:30pm		
Contract	I2309	Change	Change %	May. 23	Change	Change %
Closing Price	730.50	-31.00	-4.07%	108.00	-7.45	-6.45%
Vol traded ('000 lots)	83.35	3.19	3.98%	8.95	4.56	103.67%
Open positions ('000 lots)	78.94	1.86	2.41%	26.77	-0.04	-0.16%
Day Low	730.0	-27.00	-3.57%	107.40	-7.25	-6.32%
Day High	759.5	-17.00	-2.19%	115.20	-2.35	-2.00%

DRY BULK FREIGHT RATES (USD/MT)



TOTAL CHINA IRON ORE IMPORT VOLUMES (MILLION TONNES)



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STEEL SPOT MARKET PRICES—CHINA

Steel Spot Market RMB/tonne

Product	2023/4/21	Change	Change %
ReBar HRB400 ϕ18mm	3,970	-90	-2.22%
Wirerod Q300 ϕ6.5mm	4,180	-90	-2.11%
HRC Q235/SS400 5.5mm*1500*C	4,140	-60	-1.43%
CRC SPCC/ST12 1.0mm*1250*2500	4,610	-60	-1.28%
Medium & Heavy Plate Q235B 20mm	4,350	-30	-0.68%
GI ST02Z 1.0mm*1000*C	4,990	-10	-0.20%
Colour Coated Plate	7,400	-50	-0.67%

CHINESE STEEL MILL PROFITABILITY

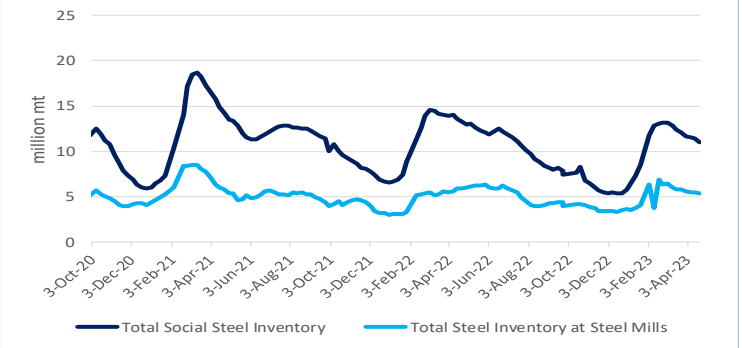
SMM Tracking of Steel Mill P&L - Rebar and Hot-rolled Coil (RMB/tonne)

Category	Price	Change (WoW)	Note
MMi (Fe 62%), USD/mt exluding tax	145.75	-1	Mmi CFR Equivalent index for 1st Feb
Coke	2,710	-100	2nd grade met coke, Tangshan, incl. tax
Steel Scrap	3,020	-50	steel scrap (6mm) in Zhangjiagang, exl. tax
Billet Cost	3,478	-59	Q234, incl. tax
Rebar cost - Blast furnace	3,964	-100	calculated based on theoretical weight, incl. tax
Rebar profit - Blast furnace	-54	-3	based on Shanghai prices, incl. tax
Hot-rolled coil cost - Blast furnace	3,977	-65	based on actual weight, incl. tax
Hot-rolled coil proft - Blast furnace	103	-65	based on Shanghai prices, incl. tax

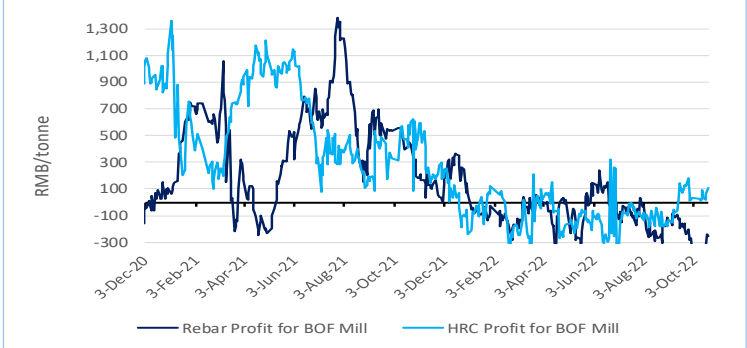
Note: 1. Costs in the table are caluclated based on todays market prices and facout our management, sales, financial and depreciations fe

2. The cost refers to average cost in the industry based on SMM's survey of small, medium and large mills in China

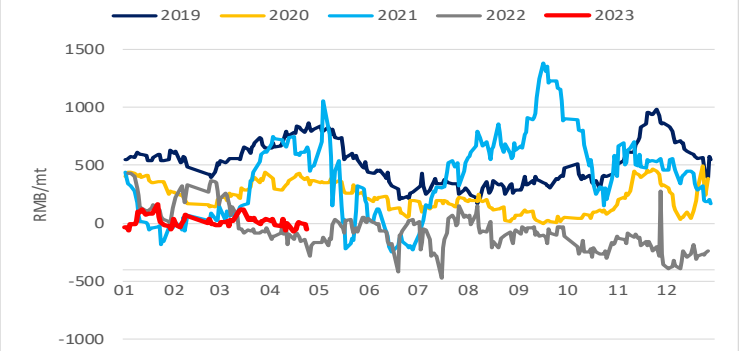
CHINESE STEEL INVENTORIES



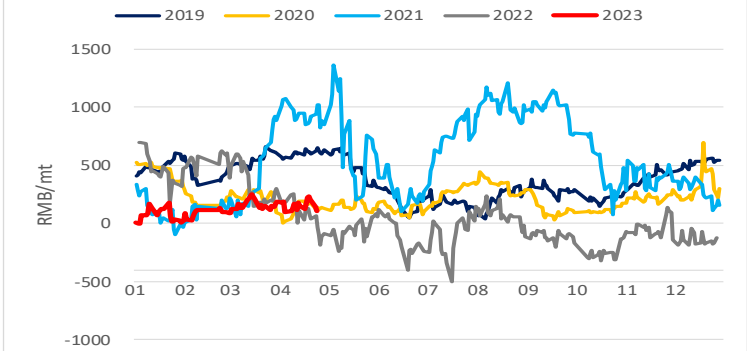
CHINESE STEEL MILL PROFITABILITY



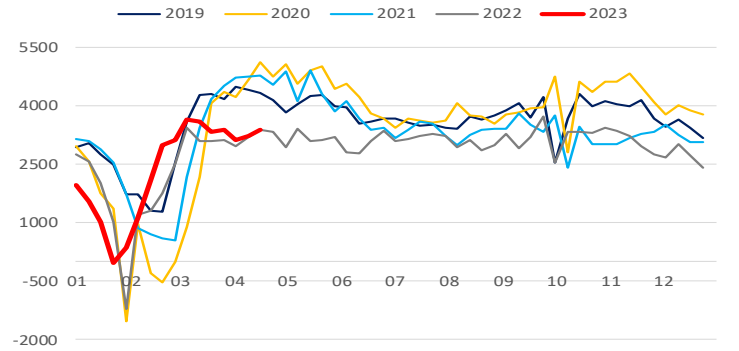
CHINESE STEEL MILL PROFITABILITY—Rebar



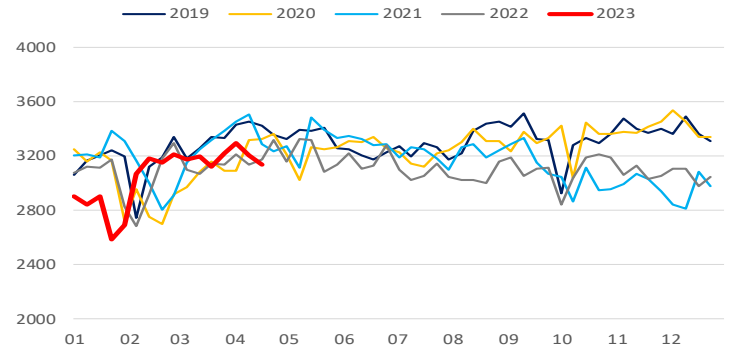
CHINESE STEEL MILL PROFITABILITY—Hot-rolled Coil



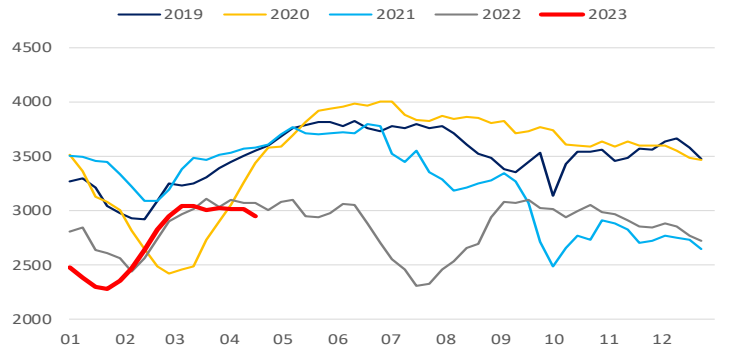
CHINESE STEEL CONSUMPTION—Rebar



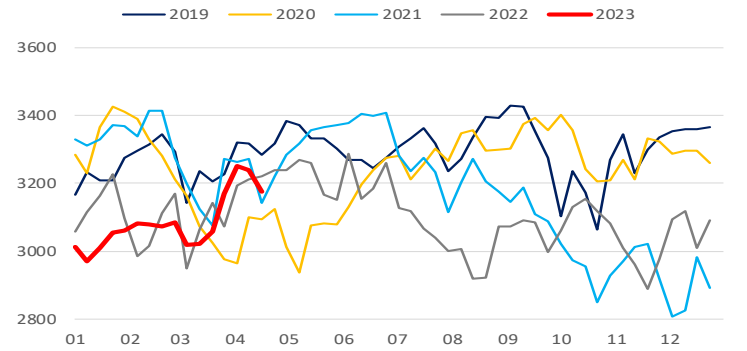
CHINESE STEEL CONSUMPTION—Hot-rolled Coil



CHINESE STEEL PRODUCTION—Rebar



CHINESE STEEL PRODUCTION—Hot-rolled Coil



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IRON ORE INDEX SPECIFICATIONS, COMPILATION RATIONALE AND DATA EXCLUSIONS

Iron Ore Index Specifications (Port and Seaborne)

	65% Fe Fines	62% Fe Fines	58% Fe Fines	62.5% Fe Lump
Fe %	65.00	62.00	58.00	62.50
Alumina %	1.40	2.25	2.25	1.50
Silica%	1.50	4.00	5.50	3.50
Phosphorus %	0.06	0.09	0.05	0.08
Sulphur %	0.01	0.02	0.02	0.02
Moisture %	8.00	8.00	9.00	4.00
Sizing	Granular size below 10mm for at least 90% of cargo; maximum of 40% below 150 micron			Size below 6.3mm max 15% Size above 31.15mm max 25%
Pricing Point	Qingdao Port (FOT and CFR respectively)			FOT Qingdao Port
Timing (Seaborne)	Loading within 4 weeks, Delivery within 8 weeks			Delivery within 2 weeks
Payment Terms	L/C at sight			L/C at sight or CAD

Iron Ore Index Compilation Rationale and Data Exclusions

MMi iron ore indices are compiled from data provided by companies that are part of the iron ore supply chain and involved in spot market transactions. The indices are calculated using detailed transaction-level data submitted to MMi by these companies. This data is normalised to the appropriate specifications and screened to remove outliers before volume-weighted average prices are calculated from the remaining core set of data.

For more details on MMi's iron ore methodology please download the guide published on our website at: www.mmiprices.com

Data Exclusions*

	62%	58%	65%
Port Index	0	0	0
Seaborne index	0		0
Lump Index 62.5	0		

* Number of price submissions for iron ore indices that were excluded from index calculations today as they were anomalous and could not be verified

IRON ORE DOMESTIC CONCENTRATE INDEX CALCULATION METHODOLOGY

The compilation method for price index generally refers to the compilation method of CPI price index and other price indices, breakdown the price data and calculate the average value according to a certain method, taking the vertical axis as the regional composite index (average of different grade index) and the horizontal axis as the grade composite index (average of different regional index), a total composite index for domestic ore can be output ultimately. The process system is also adopted in the calculation i.e. each sub-index can be obtained as well.

AVERAGE IRON ORE SPECIFICATIONS APPLIED FOR BRAND PRICE ASSESSMENTS

PORT STOCK BRANDS

Apr 21st, 2023	Specifications applied for 62% brand assessments				
	Fe	Alumina	Silica	Phos	Moisture
Roy Hill	60.88%	2.35%	4.45%	0.055%	9.26%
SIMEC Fines	60.00%	2.30%	6.30%	0.060%	6.00%
PB Fines	61.64%	2.31%	3.87%	0.099%	9.45%
Newman Fines	62.49%	2.31%	3.98%	0.090%	7.72%
MAC Fines	60.43%	2.38%	4.64%	0.078%	7.57%
Jimblebar Blended Fines	60.87%	2.95%	4.53%	0.111%	7.32%
Carajas Fines	64.90%	1.39%	1.82%	0.080%	7.78%
Brazilian SSF	62.00%	1.00%	6.50%	0.040%	6.00%
Brazilian Blend Fines	62.56%	1.58%	4.79%	0.091%	8.72%
RTX Fines	61.00%	3.10%	4.50%	0.135%	7.50%
West Pilbara Fines	60.10%	2.30%	4.70%	0.075%	8.50%

Apr 21st, 2023	Specifications applied for 58% brand assessments				
	Fe	Alumina	Silica	Phos	Moisture
SSF	56.49%	3.20%	6.19%	0.065%	9.18%
FMG Blended Fines	58.20%	2.43%	5.54%	0.057%	8.29%
Robe River	56.44%	3.16%	5.73%	0.042%	8.44%
Western Fines	57.88%	2.87%	7.50%	0.062%	7.45%
Atlas Fines	58.00%	1.85%	5.50%	0.090%	9.00%
Yandi	56.87%	1.58%	6.41%	0.042%	9.53%

SEABORNE BRANDS

	Specifications applied for 62% brand assessments				
	Fe	Alumina	Silica	Phos	Moisture
Roy Hill	60.70%	2.30%	4.90%	0.055%	8.00%
SIMEC Fines	60.00%	2.30%	6.30%	0.060%	6.00%
PB Fines 62%	62.00%	2.60%	4.30%	0.090%	9.50%
Newman Fines	62.30%	2.40%	4.30%	0.095%	8.00%
MAC Fines	60.80%	2.35%	4.40%	0.075%	7.80%
Jimblebar Blended Fines	60.50%	3.10%	4.80%	0.120%	7.80%
Carajas Fines	65.10%	1.50%	1.70%	0.080%	8.50%
Brazilian SSF	62.00%	1.00%	6.50%	0.040%	6.00%
Brazilian Blend Fines	63.00%	1.50%	5.00%	0.070%	7.00%
RTX Fines	61.00%	3.10%	4.50%	0.135%	7.50%
West Pilbara Fines	60.10%	2.30%	4.70%	0.075%	8.50%

BLOOMBERG TICKERS

PORT STOCK INDICES

	FOT Qingdao (RMB/wet tonne)	CFR Qingdao Equivalent (USD/dry tonne)
IOPI62	IRCNQ001	IRCNQ004
IOPI58	IRCNQ002	IRCNQ005
IOPI65	IRCNQ003	IRCNQ006
IOPLI62	IRCN0036	IRCN0037

SEABORNE INDICES

	CFR Qingdao (USD/DMT)
IOSI62	IRCN0034
IOSI65	IRCN0035

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